DNA, genes and chromosomes

Today I can… I can label DNA, a gene and a chromosome in a simple model. I can explain that a gene is a section of DNA with a protein instruction. I can distinguish one chromosome from the genome.

Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

TA brief · before and during

Read the exact goal and criteria. Two learners: one explains a label, one asks which part it names; swap, keeping each response. Actual staffing determines who supports; no pupil phones, public cold calling, required speech or personal/family disclosure. Book-first or agreed response mode.

Prepare enlarged full model beside unlabelled model, four word cards, blank response books and chosen input. Cognition: fade gene only. Communication: rehearse “part of”, point/sign/AAC. Sensory: high contrast/easy grip/digital picture/speech typing. Language: meanings/examples and any-language planning; keep the science demand, EAL is not SEN.

Optional kit: short cord, large blunt clips, genome/chromosome frames and tray. Clip pinch/cord catch: large blunt clips, short cord on tray away from necks. Short length only, counted out and back in, adult-supervised and removed immediately if dysregulated. Confirm school risk assessment. Offer picture alternative; no forced handling.

Arrival 5 private predictable notice. Starter 3: hear both attempts. I do 5: short DNA section → gene, packaged DNA → chromosome, all DNA → genome. Clip marks DNA; it is not the gene itself. Source models are not to scale and make no personal medical claim.

We do 7: missing label gene, match keys gene / DNA / chromosome / genome in card order; option order DNA / chromosome / gene / genome. Reset minute is inside 7. We do 2 5: correct B/second, gene within DNA. Hear both own reasons; teacher re-models an unsure learner and asks parallel genome hinge while the other continues.

Independent 8: accurate Model A plus reason and ONE chosen route. Questioner/explainer swap; no filled answer. Source two-diagram audit repairs DNA-inside-gene and one-chromosome-entire-genome. Reset minute inside 8. Review 4 includes reset, actual feedback, true edit // and requested read-back ?. Exit 3 hears each model/reason and precise sentence.

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TA brief · prompts, regulation and handoff

Held five-step prompt ladder · staff only

1. Point to the relevant place / instruction.
2. Ask one short question: "What is it asking you to find?"
3. Rephrase the instruction in simpler language.
4. Provide a scaffold or limited choice.
5. Model the first step only, then return the task to the pupil.

Wait five seconds first. Source cues: point at marked string; “Is this marker the entire string or part of it?”; model DNA on one string, learner labels coloured part; if needed complete one label together then fresh diagram alone. No physical prompts. I do: point short section, trace whole string. We do: where/what size is missing label? Independent: show incorrect label then name the short section.

Check in privately before hinge/top rung/Independent. Give heads-up for faulty-model critique, pause/space/one-label return and private correction. Follow each individual behaviour/support plan. Re-enter last completed step with one small choice. No evidence during distress.

Use actual VF · WS · I · NS(+) · E · R · // · ? only. R is next lesson’s judged response, never automatic. Record deepest prompt separately from I · S · C, valid mode and what was independent. Evidence is labelled explanation or observation/result table, Review edit and Exit beside original work; no invented data table. Use current SCI-LAUNCH-A2-W04 for spoken/practical/photo evidence on [SYS]. No pupil data in HTML.

Targets: missing-step sequencing/attention/independence; agreed hinge mode/turn-taking without mandatory speech; Arrival/pause engagement; accessible Independent input and independence; four-word language rehearsal. Link actual E evidence to personal education/support/health/care/personal targets on [SYS], not diagnosis.

Handoff: criteria met, hinge split/misconception, actual code, deepest prompt, response mode, independence, adaptation/reason, coverage change and banked evidence. Next hide genome label and reconstruct whole DNA meaning before “Complete a genetic cross (digital Punnett square).” If behind preserve individual accurate-model choice/reason/safety; teacher may shorten secure rehearsal or move Exit reflection to Week 5 Arrival with a record. Date/Version _____.

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